

# Paper III - Subspeciality of Urology (practice paper, MCh Urology Summer 2015 set)

## Contents

1. III-1. Ureteric stents; ureteroscopy with or without stenting (LAQ)
2. III-2. Medical management of VUR; 8-yr boy with bilateral grade IV VUR (LAQ)
3. III-3a. Sacral neuromodulation
4. III-3b. Uses of buccal mucosa in urology
5. III-3c. HSG (USG) in neonates
6. III-3d. Penile prosthesis types and biofilms
7. III-3e. Uretero-vaginal fistula
8. III-3f. Prune belly syndrome

**III-1. Discuss various ureteric stents. Discuss ureteroscopy with or without stenting.**

**Type:** LAQ (25 marks)

## Introduction

- Ureteric stent = hollow tube keeping the upper tract draining past obstruction, injury or anastomosis.
- **Zimskind 1967**: first indwelling silicone splint (migrated). **Finney 1978: double-J**; self-retaining coils solved migration.

## Stent design

| Parameter | Details                                                                                                                                     |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Diameter  | 3-8.5 Fr (adults 4.7-7 Fr; 6 Fr default); bigger → better extraluminal flow                                                                 |
| Length    | Adults 20-30 cm (usual 24-26 cm), from height or measured on ureteric catheter; children ≈ age + 10 cm                                      |
| Coils     | Double-pigtail, J, multi-length, loop-tail (fewer symptoms)                                                                                 |
| Features  | Radio-opaque, hydrophilic coating, extraction string                                                                                        |
| Flow      | Mostly <b>extraluminal</b> + side holes; poor in malignant extrinsic compression → tandem/metallic stents or PCN (percutaneous nephrostomy) |

**Diagram to draw:** double-J in situ (renal coil, side holes, bladder coil).

## Materials

| Material                       | Features                                              |
|--------------------------------|-------------------------------------------------------|
| Polyethylene                   | Rigid, brittle; abandoned                             |
| Silicone                       | Most biocompatible, least encrustation; high friction |
| Polyurethane                   | Stiff, cheap; more encrustation                       |
| Copolymers (Percuflex, C-Flex) | Soften at body temperature                            |
| <b>Metallic - Resonance</b>    | Malignant obstruction; up to 12 months                |

| Material            | Features                                                                                                                       |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Memokath 051</b> | Thermo-expandable nitinol; expands with warm saline, cooling softens for removal; long-term strictures                         |
| Coated              | <b>Heparin</b> (anti-encrustation), <b>triclosan</b> (no convincing benefit), hydrogel, silver, drug-eluting (investigational) |
| Biodegradable       | No removal; experimental (fragmentation)                                                                                       |

- Special: endopyelotomy (tapered 14/7 Fr), magnetic-tip, stent on string (no cystoscopy; dislodgement), mono-J (diversion), anti-reflux valve.

## Indications

- **Obstruction**: stone with sepsis (stent = PCN), malignancy, stricture, pregnancy.
- **After intervention**: URS (ureteroscopy, selected), endopyelotomy, pyeloplasty, reimplant, transplant.
- **Healing**: injury, leak, fistula. **Prophylactic**: complex pelvic surgery, passive dilatation.

## Stent-related symptoms

- Up to ~80%: frequency, urgency, flank pain on voiding (reflux), haematuria; **USSQ (Ureteral Stent Symptom Questionnaire; Joshi 2003)**.
- Mechanism: trigonal irritation (coil crossing midline), reflux, ureteric spasm.
- Rx: **alpha-blocker + antimuscarinic** (combination > either); correct length, soft/tail stents, shortest dwell.

## Complications

- Migration (proximal → URS retrieval; distal → symptoms/expulsion), infection/biofilm, haematuria (rare uretero-arterial fistula), fracture, knotting, erosion, VUR (vesicoureteral reflux).
- **Encrustation** ↑ with time (>6 weeks–3 months); stone formers, urease UTI (urinary tract infection; Proteus), pregnancy, CKD (chronic kidney disease).
- **Forgotten stent** → obstruction, sepsis, renal loss; registries, SMS reminders.
- **FECal (Forgotten, Encrusted, Calcified) grading** (Acosta-Miranda 2009): I minimal linear; II circular; III partial coil(s); IV one coil complete; V both coils complete.

## Encrusted stent

- Never pull forcefully. NCCT (non-contrast computed tomography) for burden; DMSA (dimercaptosuccinic acid) scan if function doubtful; treat UTI.
- **Bladder coil** (cystolithotripsy) → **ureter** (URS laser) → **renal coil** (fURS (flexible URS) or PCNL (percutaneous nephrolithotomy)); ESWL (extracorporeal shock wave lithotripsy) adjunct; single session where possible; nephrectomy if non-functioning. Metabolic evaluation.

## Ureteroscopy

|            | Semirigid            | Flexible                   |
|------------|----------------------|----------------------------|
| Size       | ~6–8.5 Fr            | ~7.5–9.5 Fr                |
| Best for   | Distal/mid ureter    | Upper ureter, kidney       |
| Deflection | None                 | ~270° both ways            |
| Optics     | Rod lens/fibre-optic | <b>Digital</b> chip-on-tip |

- **UAS (ureteral access sheath)** 10/12–12/14 Fr: ↓ intrarenal pressure, better vision, repeated passage; risk **wall injury** (Traxer 2013; ~1 in 8 significant); pre-stenting ↓ injury.

- **Ho:YAG (holmium:YAG) 2120 nm** gold standard; **TFL (thulium fibre laser) ~1940 nm**: finer dust, less retropulsion.
- Dusting (faster, no UAS, residual fines) vs fragmentation + basketing (stone-free confirmed, analysis).
- Steps: sterile urine, antibiotics → safety guidewire → semirigid inspection → UAS under fluoroscopy → low-pressure irrigation, laser/basket → inspect ureter (PULS: Post-Ureteroscopic Lesion Scale) → stent decision.

## Stenting after URS

---

- **AUA (American Urological Association) 2016**: may omit after **uncomplicated** URS.
- **EAU (European Association of Urology) 2024**: do not routinely stent after uncomplicated URS; pre-stenting not routine but ↑SFR (stone-free rate).
- RCTs (randomised controlled trials)/meta-analyses: no stent → less pain/LUTS (lower urinary tract symptoms), cost; unplanned visits not significantly ↑.
- **Stent if**: injury/perforation, significant residual fragments, solitary/transplant kidney, impacted stone/oedema, pregnancy, prolonged procedure, UTI/sepsis, renal insufficiency, bilateral URS, stricture, planned second-look.
- Duration: uncomplicated 3–7 days (string); injury 2–6 weeks; stricture/endopyelotomy 4–6 weeks.

After URS -> uncomplicated?

Yes -> no stent (or string stent 3-7 d)

No -> double-J: injury 2-6 wk; residual/impacted 1-2 wk

## Recent advances

---

- **FANS (flexible and navigable suction access sheath)**: ↓intrarenal pressure, better SFR, fewer infections.
  - TFL and pulse-modulated high-power Ho:YAG.
  - **Single-use digital flexible ureteroscopes**.
-

## III-2. Describe medical management of VUR. Describe treatment of an 8-yr male child with b/l grade IV VUR.

**Type:** LAQ (25 marks)

### Definition

- **VUR (vesicoureteral reflux):** backflow of urine from bladder into ureter/upper tract.
- **Primary:** congenital UVJ (ureterovesical junction) flap-valve deficiency; short tunnel (normal length : diameter ~5:1, Paquin).
- **Secondary:** PUV (posterior urethral valves), neurogenic bladder, BBD (bladder and bowel dysfunction), iatrogenic.
- ~1% normal children, ~30-40% with febrile UTI (urinary tract infection), ~30% siblings.
- Reflux + infection → pyelonephritis; intrarenal reflux (compound papillae) → **scarring** → hypertension, CKD (chronic kidney disease) = **reflux nephropathy**. Congenital dysplasia esp. boys with high-grade antenatal VUR.

### IRS (International Reflux Study) grading (VCUG: voiding cystourethrogram)

| Grade | Findings                                                                |
|-------|-------------------------------------------------------------------------|
| I     | Ureter only                                                             |
| II    | Ureter, pelvis, calyces; no dilatation                                  |
| III   | Mild-moderate dilatation; minimal fornix blunting                       |
| IV    | Moderate dilatation/tortuosity; <b>papillary impressions maintained</b> |
| V     | Gross dilatation; <b>papillary impressions lost</b>                     |

**Diagram to draw:** grades I-V sketches; Cohen reimplant.

### Natural history

- Depends on grade, laterality, age, BBD.
- I-II ~80%+ resolve over ~5 years; III ~50%; IV unilateral ~50-60% at 5 years, **bilateral ~10%**; V rarely; mostly before puberty.

### Evaluation

- Urine culture, creatinine, BP, proteinuria.
- **RBUS (renal-bladder ultrasound)**, PVR (post-void residual).
- **VCUG:** grade, urethra (exclude PUV), duplex. Radionuclide cystogram/ceVUS (contrast-enhanced voiding urosonography) for follow-up.
- **DMSA (dimercaptosuccinic acid scan):** scars, DRF (differential renal function).
- Uroflow ± urodynamics if BBD suspected.

### Medical management

**1. CAP (continuous antibiotic prophylaxis)** - nightly, ~ $\frac{1}{4}$  therapeutic dose

| Drug                                    | Dose               | Notes                         |
|-----------------------------------------|--------------------|-------------------------------|
| TMP-SMX (trimethoprim-sulfamethoxazole) | <b>2 mg/kg TMP</b> | Avoid <2 months (kernicterus) |
| Nitrofurantoin                          | <b>1-2 mg/kg</b>   | Avoid CKD/G6PD deficiency     |
| Cephalexin                              | ~10 mg/kg          | Infants <2 months             |

## 2. Evidence

- **RIVUR (NEJM 2014)**: 607 children, 2–71 months, grade I–IV; TMP-SMX vs placebo → **recurrent UTI halved**, most benefit with **BBD**; **no difference in new scarring**; more resistant organisms.
- **Swedish Reflux Trial (Brandström 2010)**: 1–2 yr, grade III–IV; CAP vs endoscopic vs surveillance → CAP/endoscopy ↓ febrile UTI in **girls**, more scars on surveillance; **boys** few recurrences in any arm.

**3. BBD treatment** first: timed/double voiding, treat constipation, urotherapy/biofeedback, anticholinergics, CIC (clean intermittent catheterisation) if needed. BBD ↓ resolution and endoscopic success.

**4. Circumcision**: ↓ UTI in infant boys.

**5. Surveillance without CAP**: low-grade, no BBD, older child.

**6. Follow-up**: RBUS 6–12 months; VCUG 12–24 months; DMSA if breakthrough.

## Guidelines

---

- **AUA (American Urological Association) 2010**: CAP recommended with **BBD + VUR**; treat BBD first; breakthrough febrile UTI → consider intervention.
- **EAU/ESPU (European Association of Urology/European Society for Paediatric Urology)**: **high-risk** (grade IV–V, scars, BBD) → CAP, **early intervention** if breakthrough; low-risk → surveillance or CAP. Reimplant for high grade, endoscopic for low-moderate.

**Indications for intervention**: breakthrough febrile UTI; new scars/↓ DRF; persistent IV–V; CAP non-compliance; paraureteric diverticulum/duplex; parental preference.

## Case: 8-yr boy, bilateral grade IV VUR

---

### Assessment

- Exclude **PUV** (VCUG urethra), neurogenic bladder (spine), **BBD** (diary, uroflow, PVR).
- UTI history; DMSA, creatinine, BP.
- Bilateral IV at 8 → **resolution ~10%**.

### Options

1. **CAP + BBD management**: if no febrile UTI, no scars; boys past infancy have lower UTI risk. Cons: resistance, compliance.
2. **Endoscopic Deflux (dextranomer/hyaluronic acid)**: STING (subureteric) or HIT (hydrodistension-implantation). **Elder 2006** one-injection success: **IV ~63%** (I–II ~79%, V ~51%); ↑ with repeat. Day-care; lower durability.
3. **Ureteral reimplantation** (gold standard, ~**95–98%**):
  - **Cohen cross-trigonal**: commonest; later URS (ureteroscopy) access difficult.
  - **Extravesical Lich-Gregoir**: less haematuria/spasm; **bilateral → transient retention risk** (pelvic plexus), ↓ with nerve-sparing.
  - **RALUR (robotic-assisted laparoscopic ureteral reimplantation)**: less pain, shorter stay; ~85–95%.
  - **Bilateral Cohen** ideal: one bladder opening corrects both.

### Plan

- Shared decision; treat BBD before surgery.
- **Breakthrough UTI, scars, DRF loss → bilateral reimplant**; Deflux alternative (~60–70% per session).
- **Asymptomatic, no scars** → CAP + BBD therapy; intervene if UTI/new scar.
- Postop: CAP ~3 months; RBUS 4–6 weeks; VCUG optional after open, usually after Deflux; long-term BP, proteinuria.

Bilateral grade IV VUR, 8 yr -> exclude PUV, treat BBD  
-> UTI / scars / DRF loss -> Cohen (~95-98%) | RALUR | Deflux (~63%)  
-> none -> CAP + BBD rx -> intervene if UTI / new scar

## Recent advances

---

- ceVUS avoids radiation; "top-down" (DMSA first) approach.
  - Selective CAP after RIVUR.
-

### III-3a. Sacral neuromodulation

**Type:** SAQ (10 marks)

#### Definition and mechanism

- **SNM (sacral neuromodulation):** continuous low-amplitude stimulation of **S3 root** via tined lead + IPG (implantable pulse generator).
- **Afferent modulation** (not motor) at spinal/supraspinal levels → works in **both overactivity and retention** (suppresses sphincter "guarding reflex").

#### Indications (FDA: Food and Drug Administration)

| Indication                                                         | FDA  |
|--------------------------------------------------------------------|------|
| Refractory UUI (urgency urinary incontinence)                      | 1997 |
| Urgency-frequency                                                  | 1999 |
| <b>Non-obstructive retention</b> (incl. <b>Fowler's syndrome</b> ) | 1999 |
| Faecal incontinence                                                | 2011 |

- **Fowler's:** young women, retention, abnormal sphincter EMG (electromyography); SNM best treatment.

#### Technique

1. **Test phase:** PNE (percutaneous nerve evaluation) 3–7 days (false negatives) vs **staged tined lead** ~1–2 weeks (more reliable).
2. Prone; needle into **S3 foramen** (≈9 cm above coccyx tip, ~2 cm lateral).
3. **S3 response: bellows + big-toe plantar flexion.** S2: anal clamp, whole-foot. S4: bellows only.
4. **≥50% improvement** in diary → stage 2 IPG in upper buttock.

#### Outcomes

- **InSite (Siegel 2015):** SNM > standard medical therapy at 6 months; durable at 5 years.
- **ROSETTA (JAMA 2016):** Botox 200 U vs SNM, refractory UUI → Botox slightly greater ↓ UUI at 6 months, but **more UTIs (urinary tract infections) and CISC (clean intermittent self-catheterisation)**; SNM had revisions.

#### Complications

- IPG-site pain, lead migration, loss of efficacy, infection (explant). Revision historically ~30%+.

#### Contraindications

- Failed test, cannot use programmer, sacral anomaly, repeated MRI (magnetic resonance imaging) with old devices; relative: pregnancy.

#### Recent advances

- Rechargeable/recharge-free full-body MRI-conditional IPGs; tibial nerve stimulation (PTNS: percutaneous tibial nerve stimulation; implantable).

**Diagram to draw:** sacrum with S3 lead + buttock IPG.

-----

### III-3b. Uses of buccal mucosa in urology

Type: SAQ (10 marks)

#### History

- **BMG (buccal mucosal graft)**: workhorse free graft for urethroplasty.
- **Humby 1941** (hypospadias); **Bürger 1992/El-Kasaby 1993** (stricture); Morey & McAninch 1996 (ventral onlay); Barbagli 1996 (dorsal onlay).

#### Advantages

- Thick **non-keratinised** epithelium, infection-resistant.
- **Thin, vascular lamina propria + panlamina plexus** → excellent take.
- Hairless, wet-adapted, easy harvest, hidden scar; usable in lichen sclerosus.

#### Harvest

- Nasotracheal intubation; ~4–6 × 2–2.5 cm per cheek.
- Stay **1 cm from commissure**; **avoid Stensen's duct opposite upper 2nd molar**.
- Infiltrate saline ± adrenaline, excise mucosa without muscle, haemostasis; close or leave open (evidence mixed); **defat** on back table.
- Alternative: **lingual graft** (Simonato 2006).
- Donor morbidity (mostly transient): pain, numbness, trismus, duct injury, bleeding.

#### Uses

| Area                                  | Use                                                                                                                                                                                                                                                                               |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Anterior urethroplasty</b>         | Bulbar: dorsal onlay (Barbagli), ventral onlay (Morey), <b>dorsal inlay (Asopa 2001)</b> , <b>Kulkarni one-sided dorsal</b> (2009; preserves ventral blood supply, bulbospongiosus), augmented anastomotic (excise obliterative segment + graft rest). Penile: dorsal inlay/onlay |
| <b>Staged</b> (Johanson, Bracka)      | Lay open + graft → tubularise ~6 months; <b>lichen sclerosus</b> , failed hypospadias, panurethral                                                                                                                                                                                |
| Hypospadias                           | Redo/"cripples"                                                                                                                                                                                                                                                                   |
| Bladder neck/vesico-urethral stenosis | Robotic BMG onlay (emerging)                                                                                                                                                                                                                                                      |
| <b>Ureter</b>                         | <b>Robotic BMG ureteroplasty (Zhao)</b> + omental wrap for long proximal/mid strictures; avoids ileal ureter                                                                                                                                                                      |
| Others                                | Vaginoplasty, female urethral stricture, Peyronie's (rarely), neourethra in exstrophy (selected)                                                                                                                                                                                  |

#### Outcomes

- Anterior urethroplasty ~**85-90%** (dorsal = ventral in bulbar); lower in penile, lichen sclerosus, redo.
- Robotic ureteroplasty ~90% short-term.

#### Recent advances

- Tissue-engineered oral mucosa (investigational); robotic BMG for ureter/bladder neck; single-cheek harvest without closure.

**Diagram to draw:** bulbar cross-section: dorsal onlay, ventral onlay, Asopa inlay.

-----

### III-3c. HSG in neonates

**Type:** SAQ (10 marks)

*HSG (hysterosalpingography) is not done in neonates → taken as "USG (ultrasonography) in neonates".*

#### Introduction and timing

- First-line, radiation-free; mainly **ANH (antenatal hydronephrosis)**, ~1-2% of pregnancies.
- Also: febrile UTI (urinary tract infection), mass, oligohydramnios, poor stream.
- **Avoid first 48-72 h** (oliguria underestimates dilatation); **within 24-48 h** if suspected **PUV (posterior urethral valves)**, bilateral severe HDN (hydronephrosis), solitary kidney. Normal → **repeat 4-6 weeks**.
- **APD (anteroposterior diameter)**: transverse view at hilum.

#### Normal neonatal kidney

- ~4-5 cm; **prominent hypoechoic pyramids** (not hydronephrosis); **cortex iso/hyperechoic to liver**.

#### Grading

- **SFU (Society for Fetal Urology) 0-4**: 0 none; 1 pelvis; 2 + some calyces; 3 all calyces; 4 + parenchymal thinning.

#### UTD (urinary tract dilation) 2014 - postnatal (≥48 h)

| Grade                    | Findings                                                             |
|--------------------------|----------------------------------------------------------------------|
| <b>P1</b> (low)          | APD <b>10-&lt;15 mm</b> or central calyces                           |
| <b>P2</b> (intermediate) | APD <b>≥15 mm</b> , peripheral calyces or <b>ureteral dilatation</b> |
| <b>P3</b> (high)         | P2 + <b>abnormal parenchyma</b> or <b>bladder</b>                    |

#### Specific conditions

| Condition                                             | USG findings                                                                                                                     |
|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| PUJO (pelviureteric junction obstruction)             | Pelviccalyceal dilatation, normal ureter/bladder                                                                                 |
| VUR (vesicoureteral reflux)                           | Fluctuating dilatation; <b>normal USG does not exclude</b>                                                                       |
| <b>PUV</b>                                            | Bilateral HUN (hydroureteronephrosis), <b>thick bladder; keyhole sign antenatal</b> ; confirm by VCUG (voiding cystourethrogram) |
| MCDK (multicystic dysplastic kidney)                  | <b>Non-communicating</b> cysts                                                                                                   |
| ARPKD (autosomal recessive polycystic kidney disease) | Bilateral <b>enlarged echogenic</b> kidneys                                                                                      |
| Adrenal haemorrhage                                   | <b>Avascular</b> suprarenal mass                                                                                                 |

#### Follow-up

Postnatal USG (day 3-7; 24-48 h if PUV/bilateral)  
Normal → repeat 4-6 wk; P1 → USG 1-6 mo  
P2 → USG 1-3 mo +/- VCUG, isotope renogram, prophylaxis  
P3 → USG 1 mo, VCUG, isotope scans, prophylaxis; urgent if PUV

**Diagram to draw:** SFU 0–4 sketches; transverse APD.

---

### III-3d. Types of penile prosthesis. What are biofilms in penile prosthesis?

**Type:** SAQ (10 marks)

#### Introduction

- Third-line for ED (erectile dysfunction) refractory to medical/intracavernosal therapy; satisfaction **>85-90%**.

#### Types

| Type                                              | Examples                                                        | Pros / cons                                                                        |
|---------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------------------------------------|
| <b>Malleable</b>                                  | Genesis, Spectra, Tactra                                        | Cheap, no mechanical failure / constant rigidity                                   |
| <b>2-piece</b>                                    | <b>Ambicor</b> (pump-reservoir in scrotum)                      | No abdominal reservoir (prior pelvic surgery, transplant) / less flaccidity, girth |
| <b>3-piece IPP (inflatable penile prosthesis)</b> | <b>AMS 700 CX/LGX/CXR (narrow, fibrosis); Titan/Titan Touch</b> | Most natural / cost, mechanical failure                                            |

- IPP: cylinders, scrotal pump (lock-out valve), reservoir in **Retzius** or **ectopic submuscular** (after pelvic surgery). Mechanical survival ~80-90% at 5 years, 60-70% at 10 years.
- Coatings: **InhibiZone** = rifampin + minocycline; **Coloplast hydrophilic** dipped in antibiotic → infection ~halved.

#### Biofilm

- **Definition:** structured microbial community in a self-produced **EPS (extracellular polymeric substance)** matrix adherent to the implant; persister cells.
- **Stages:** conditioning film → reversible/irreversible adhesion → microcolonies → maturation (quorum sensing) → dispersal (planktonic seeding) → clinical infection.
- **Organisms:** **CoNS (coagulase-negative staphylococci)**, **S. epidermidis** commonest; S. aureus, Gram-negatives, anaerobes, Candida (diabetics); often polymicrobial.
- **Antibiotic failure:** poor penetration, dormant cells; **MIC (minimum inhibitory concentration)** **100-1000x** higher.

#### Infection

- **~1-3% virgin**; higher in revision, poorly controlled diabetes, SCI (spinal cord injury), immunosuppression.
- **Prevention:** preop UTI (urinary tract infection) control, chlorhexidine-alcohol, intravenous antibiotics, "**no-touch**" (**Eid**), coated device, irrigation, short surgery.
- **Mulcahy salvage (1996):** remove all components → **7-solution washout** → immediate replacement; **~80-90%**. Not with sepsis, necrosis, erosion, DKA (diabetic ketoacidosis).
- Otherwise explant ± malleable placeholder.

#### Recent advances

- Tactra malleable (2019), Rigicon Infla10 IPP; ectopic reservoir standard after pelvic surgery.

**Diagram to draw:** 3-piece IPP in situ; biofilm life cycle.

-----

### III-3e. Uretero-vaginal fistula

**Type:** SAQ (10 marks)

## Definition and aetiology

- **UVF (uretero-vaginal fistula):** ureter-vagina communication → continuous leak per vaginum.
- **Hysterectomy** (abdominal > laparoscopic/vaginal): commonest in developed countries; injury at uterine artery crossing, cardinal ligament, vaginal cuff.
- **Obstetric** (caesarean, obstructed labour): common in developing countries incl. India.
- Others: malignancy, **radiation**, colorectal surgery.

## Presentation

- **Continuous leak + normal voiding, 1-4 weeks** post-surgery; flank pain, fever, urinoma.

## Investigations

- **Three-swab test** (bladder methylene blue): blue = VVF (vesicovaginal fistula); **wet, clear** = **UVF**; oral phenazopyridine confirms ureteric source.
- **CTU (computed tomography urography)**: extravasation, hydroureteronephrosis, urinoma.
- **Cystoscopy + RGP (retrograde pyelogram)**: site, wire passage, stent same sitting.
- **Exclude coexisting VVF (~10-15%)**, bilateral injury. Creatinine, culture.

## Management

**Principles:** drain kidney, preserve function, treat sepsis, exclude malignancy.

### 1. Endourological (first line if continuity preserved)

- **Retrograde double-J** if wire crosses; else **PCN (percutaneous nephrostomy) + antegrade** or **rendezvous**.
- Stent ~6-8 weeks → RGP/CT; many heal; stricture → dilatation.

## 2. Surgical (stent failure, transection, ligation, stricture)

- Distal (juxtavesical) → **ureteroneocystostomy: psoas hitch** (~5 cm), **Boari flap** (~10-15 cm).
- Long defects: TUU (transureteroureterostomy), ileal ureter, autotransplant.
- **Early repair** (first weeks, stable, no sepsis) now preferred over 3-6 month wait; delay if infection, radiation, poor condition.
- Open, laparoscopic or robotic.

- Leak + normal voiding -> 3-swab: clear wet = UVF
- > CTU + cysto/RGP (exclude VVF, bilateral)
- > wire passes? Yes -> double-J 6-8 wk
- No -> PCN +/- antegrade/rendezvous
- > fails -> reimplant + psoas hitch/Boari, early if fit

## Prevention / advances

- Identify ureter at hysterectomy; selective prophylactic stents; ICG (indocyanine green) fluorescence; robotic reimplant.

**Diagram to draw:** three-swab test; psoas hitch.

### III-3f. Prune belly syndrome

**Type:** SAQ (10 marks)

#### Definition

- **Eagle-Barrett/triad syndrome:** (1) deficient abdominal musculature; (2) urinary tract dilatation; (3) bilateral intra-abdominal undescended testes.
- ~1 in 30,000–50,000 live births; **~95% male**.

#### Pathogenesis

- **Mesodermal arrest** (~6–10 weeks) vs **urethral obstruction theory** (fetal outlet obstruction → bladder distension).
- Mostly sporadic; **CHRM3** in familial cases.

#### Features

- **Kidneys:** dysplasia (determines prognosis). **Ureters:** dilated, tortuous; VUR (vesicoureteral reflux) ~75%.
- **Bladder:** large, smooth, poor contractility, large PVR (post-void residual), urachal pseudodiverticulum.
- **Prostatic hypoplasia:** dilated triangular prostatic urethra; ± megalourethra.
- **Testes:** intra-abdominal; infertility, malignancy risk.
- Extra-GU (genitourinary): pulmonary hypoplasia; cardiac ~10%; orthopaedic ~45%.

#### Woodard classification

| Category | Features                                                             | Outlook                                     |
|----------|----------------------------------------------------------------------|---------------------------------------------|
| I        | Oligohydramnios, <b>pulmonary hypoplasia</b> , renal dysplasia       | Fatal                                       |
| II       | Full triad, moderate-severe uropathy, <b>no pulmonary hypoplasia</b> | Urosepsis/CKD (chronic kidney disease) risk |
| III      | Mild/incomplete, <b>stable renal function</b>                        | Good                                        |

#### Management

- **Neonatal:** respiratory support; creatinine (nadir prognostic); USG (ultrasonography), **VCUG (voiding cystourethrogram) with caution**, DMSA (dimercaptosuccinic acid)/MAG3 (mercaptoacetyl triglycine) scans; CAP (continuous antibiotic prophylaxis); avoid instrumentation.
- **Conservative** for most: CAP, timed/double voiding, CIC (clean intermittent catheterisation).
- **Drainage** (infection, rising creatinine, obstruction): **cutaneous vesicostomy** preferred; pyelostomy/ureterostomy for upper tract.
- **Reconstruction** (selected): reduction cystoplasty, ureteral tailoring + reimplant, Mitrofanoff.
- **Orchiopexy:** **transabdominal, by 6–12 months**; Fowler-Stephens for short vessels; combine with abdominoplasty.
- **Abdominoplasty:** Randolph, Ehrlich, **Monfort** (most popular).
- **Prognosis:** ~one-third CKD; transplant results similar to other causes.

#### Variants

- Females ~5% (no testicular component); "pseudo-prune belly" = incomplete triad, milder.

- Recent: fetal vesicoamniotic shunting (benefit unproven).

**Diagram to draw:** VCUG – large bladder, urachal diverticulum, triangular prostatic urethra.

---

## Rapid Revision Sheet - Paper III

*All the key points from this paper in one place. Read this in the last 30 minutes before the exam.*

### **III-1. Discuss various ureteric stents. Discuss ureteroscopy with or without stenting. (LAQ)**

- Finney 1978 double-J; Memokath 051 = nitinol; Resonance = metallic.
- FECal I-V; remove bladder coil → ureter → kidney.
- AUA 2016/EAU 2024: stent may be omitted after uncomplicated URS.
- Ho:YAG 2120 nm vs TFL ~1940 nm.

### **III-2. Describe medical management of VUR. Describe treatment of an 8-yr male child with b/l grade IV VUR. (LAQ)**

- IRS grade IV keeps papillary impressions, grade V loses them.
- RIVUR 2014: CAP halves recurrent UTI, no change in scarring.
- Deflux grade IV ~63% per injection; open reimplant ~95-98%.
- Bilateral extravesical → retention risk.

### **III-3a. Sacral neuromodulation (SAQ)**

- FDA: 1997 UUI; 1999 retention; 2011 faecal.
- Afferent modulation; S3 = bellows + big-toe plantar flexion.
- Test success ≥50%; staged timed lead beats PNE. Fowler's responds best.
- ROSETTA 2016: Botox slightly better, more UTIs/CISC.

### **III-3b. Uses of buccal mucosa in urology (SAQ)**

- Humby 1941; Bürger 1992/El-Kasaby 1993.
- Thin lamina propria + panlamina plexus = excellent take; avoid Stensen's duct; lingual alternative.
- Barbagli dorsal onlay, Asopa inlay, Kulkarni one-sided; staged Johanson/Bracka for lichen sclerosis.
- Urethroplasty success ~85-90%.

### **III-3c. HSG in neonates (SAQ)**

- Avoid first 48-72 h except suspected PUV/bilateral; repeat 4-6 weeks.
- Prominent hypoechoic pyramids + echogenic cortex are normal.
- UTD 2014: P1 10-15 mm; P2 ≥15 mm; P3 + parenchyma/bladder.
- Keyhole sign = antenatal sign of PUV.

### **III-3d. Types of penile prosthesis. What are biofilms in penile prosthesis? (SAQ)**

- Malleable (Genesis, Tactra), 2-piece (Ambicor), 3-piece (AMS 700, Titan).
- InhibiZone = rifampin + minocycline; Coloplast = hydrophilic dip.
- Biofilm = EPS-matrix community; *S. epidermidis* commonest; MIC up to 1000×
- Infection ~1-3% virgin; Mulcahy salvage ~80-90%.

### **III-3e. Uretero-vaginal fistula (SAQ)**

- Hysterectomy commonest; obstetric in developing countries.
- Continuous leak + normal voiding; clear wet swab = UVF.
- Exclude coexisting VVF (~10-15%).
- Stent if wire passes; else psoas hitch/Boari reimplant, early if fit.

### **III-3f. Prune belly syndrome (SAQ)**

- Triad: deficient abdominal wall, urinary dilatation, intra-abdominal testes; ~95% male.

- Woodard I fatal (pulmonary hypoplasia); II uropathy; III mild.
- Mostly conservative; vesicostomy if drainage needed.
- Orchiopexy by 6-12 months + Monfort abdominoplasty.